



Technology Studio at a Glance

Print pages 1-3 for each working adult; keep this duplex on a clipboard.

Purpose

Students are trying technologies before choosing how to use them. The finish line is understanding, not polished work. Honest 'not yet' evidence counts.

Mentor assignments

Stamp	Circuit Lab	micro:bit	NeoPixel
_____	_____	_____	_____
Robotics lead	Technical float	Final count lead	Room/time lead
_____	_____	_____	_____

Schedule

Time	Wearables	Robotics	Safety / understanding gate
9:51-10:16	Rotation 1	Challenge 1: identity + reset	Core action and evidence started
10:16-10:19	Count, power off, move	SAFE RESET; switch roles	Mentor releases transition
10:19-10:44	Rotation 2	Challenge 2: input	Student can show or explain
10:44-10:47	Count, power off, move	SAFE RESET; switch roles	No loose or powered parts
10:47-11:12	Rotation 3	Challenge 3: two outputs	Evidence saved before extension
11:12-11:15	Count, power off, move	SAFE RESET; switch roles	Inspect items labeled
11:15-11:40	Rotation 4	Challenge 4: interaction	Fourth capability check
11:40-11:45	Evidence + final count	Evidence + kit inventory	Ready / inspect / missing reported

Movement

- Wearables move clockwise with only the studio passport, iPad, and named stamp bag.
- Robotics teams stay with one numbered kit and switch Driver / Navigator roles.
- Do not open an unsupported needle or powered station. Reduce capacity or use the prepared visual backup.



Keep It Feeling Like a Studio

Short explanations, hands-on noticing, quick conversation, and safe reset.

Ask, then step back

- Show me what changed.
- Trace the path with your finger.
- What did you expect?
- What evidence supports that?
- What would you try next?
- Accept arrows, labels, gestures, oral explanations, and meaningful unfinished work.

Quick understanding check

A student understands when she can do at least one:

- SHOW the behavior.
- TRACE the path or system.
- PREDICT what will happen.
- EXPLAIN one cause-and-effect relationship.
- RESET the system safely.

Before release

- ☐ Test every example, file, cable, board, battery, circuit, and numbered kit.
- ☐ Match labels across each device set; open starter files; stage known-good swaps and photo/video backups.
- ☐ Count needles and kits; place READY, BUILD, CHECK, EVIDENCE, RETURN, and INSPECT areas.
- ☐ Give each student/team only the minimum studio sheets. Canvas uploading happens later.

Help ladder

Step	Mentor move
1	Name expected versus observed.
2	Check one variable.
3	Retest once.
4	Use a known-good swap.
5	After 60-90 seconds, label the item for inspect and continue with backup evidence.

Immediate stop conditions

STOP, MAKE SAFE, AND CALL THE TECHNICAL LEAD

Missing needle; crossed circuit paths; warmth or odor; exposed or damaged wire; unstable flicker; wrong robot moving; loose battery; forced, buzzing, stalled, or obstructed servo.



Master Route, Transition, and Final Count

Post one copy; keep one copy with the room/time lead.

Group	9:51	10:19	10:47	11:15
W1	Future Stamp	Circuit Lab	micro:bit	NeoPixel
W2	Circuit Lab	micro:bit	NeoPixel	Future Stamp
W3	micro:bit	NeoPixel	Future Stamp	Circuit Lab
W4	NeoPixel	Future Stamp	Circuit Lab	micro:bit
Robotics	Controller + reset	Read input	Test outputs	Interaction

Three-minute transition

Time	Everyone	Wearables	Robotics
0:00-0:30	Hands off; save screen	Stop stitching/building	Stop program
0:30-1:30	Make safe	Needle to tray; power off	SAFE RESET; disconnect; power off
1:30-2:00	Account + label	Count; bag stamp; inspect failures	Count kit; save evidence; inspect failures
2:00-2:40	Move or switch	Move clockwise	Kit stays; partners switch roles
2:40-3:00	Seat; hands off	Match incoming roster	Confirm new Driver / Navigator

11:40-11:45 final report

- ☐ Wearables: four evidence checks; stamp/grid/floss bagged; every needle returned separately; all power off.
- ☐ Robotics: four evidence boxes; both names and contributions; SAFE RESET; disconnect/off; full kit inventory.
- ☐ Station leads report READY / INSPECT / MISSING. Unresolved counts stop dismissal to the next block.



W1 - Future Stamp

Meaningful progress counts. Ordinary embroidery floss only.

Finish line

Connect one future need to a simple 6-12-stitch symbol and explain the meaning.

Say

Need -> symbol -> small grid. Do not rush or force it.

Before students

- ☐ Inspect all blanks; quarantine cracks, splinters, rough edges, and blocked holes.
- ☐ Pretest the exact needle, threader, floss thickness, and hole; cut short ordinary-floss lengths.
- ☐ Stage 6 x 6 planning grids, named bags, paper backups, one finished and one in-progress sample.
- ☐ Record opening needle count. No conductive thread, LEDs, or batteries at this table.

Exact 25-minute flow

Min.	Student action	Mentor gate
0-2	Name bag; review samples and needle rule	Needle is in a hand or counted tray
2-5	Name a future need; sketch small symbol	Reduce oversized ideas
5-7	Inspect blank; receive needle + floss	Quarantine damage
7-17	Stitch gently; check front/back	Stop forcing immediately
17-20	Show progress; secure/trim as directed	Accept honest in-progress work
20-23	Save photo + meaning sentence	Verify meaning is clear
23-25	Bag blank/grid/floss; return needle	Recount before release

Show me

Point to the need, symbol, and strongest stitch. Complete: 'My symbol represents ____ because ____.'

Stop

Lost needle, cracked/splintered blank, blocked hole, or any stitch that requires force.

Quick fixes

- Too large: stitch only the strongest feature.
- Tangle: let needle hang; shorten working floss.
- Blocked hole or damaged wood: stop and use a tested or paper backup.
- Time: photograph meaningful progress; bag work; return needle.



W2 - LED + Conductive-Thread Circuit Lab

Prepared powered loop + short unpowered stitch practice.

Finish line

Make a prepared LED glow, trace the complete loop, and practice one safe unpowered connection.

Say

Battery -> positive path -> LED -> negative path. Match + to + and - to -. Paths never touch.

Before students

- ☐ Test one known-good prepared circuit per pair and mark actual polarity.
- ☐ Prepare safe open/reversed and crossing/loose-tail fault samples; short-risk samples remain unpowered.
- ☐ Stage loop visual, premarked felt, pre-threaded needles, known-good swaps, inspect bin, and count tray.
- ☐ Battery changes remain mentor-controlled. Decorative floss stays at the Stamp table.

Exact 25-minute flow

Min.	Student action	Mentor gate
0-2	Holder off; name battery, LED, + and -	Count needles; state stop rules
2-5	Compare complete loop and path touch	Never power a short example
5-9	Trace both paths; brief prepared test; off	Approve steady-light test
9-16	Practice snug loops + flat stitches unpowered	Coach short tails and separation
16-19	Partner-check both sides; diagnose fault	Ask where path opens/reverses/crosses
19-22	Save photo + safety sentence	Check concept, not polish
22-25	Power off; return needle/circuit	Secure battery; recount

Show me

Trace positive and negative paths without crossing. Complete: 'I prevented a short by ____.'

Stop

Warmth, odor, damage, exposed crossing, touching paths, or a light that works only when squeezed.

Quick fixes

- Dark: switch/power, polarity, contact, then one known-good part.
- Flicker/pressure-dependent: power off; loose connection goes to inspect.
- Fuzzy bridge or paths touch: do not power; trim or rebuild.
- After 60-90 seconds: swap prepared circuit and protect learning time.



W3 - micro:bit + MakeCode

Prepared A/B/A+B starter on matched W-labeled equipment.

Finish line

Change and test one prepared program; explain button input and display output.

Say

Button is the input. The event decides. The display is the output. Simulate -> download -> wait -> test.

Before students

- ☐ Open and test the starter on every Windows laptop; stage offline files/screenshots.
- ☐ Match W board, USB data cable, battery holder, and bag. Keep every R board physically separate.
- ☐ Function-test A, B, A+B, reset, cable, port, and matched battery pair.
- ☐ Stage a known-good board, cable, laptop, and printed transfer sequence.

Exact 25-minute flow

Min.	Student action	Mentor gate
0-2	Match labels; battery off; choose roles	Verify W board + matched set
2-5	Point to A, B, display, reset, P0, GND	Name only today's parts
5-9	Test A, B, A+B in simulator	All three work before editing
9-14	Change one icon/display choice; retest	Preserve A+B clear/off
14-19	Download; wait until flashing ends	Battery disconnected; no early unplug
19-22	USB off; battery on; test + reset	Match physical behavior
22-25	Save evidence; power off; return kit	Verify file and kit label

Show me

Press one input and explain the output. Complete: 'When I press ___, the micro:bit ___.'

Stop

Battery connected during USB transfer, premature unplug, mismatched kit, damaged cable/board, or unexpected R firmware.

Quick fixes

- No MICROBIT drive: reseat both ends; try known-good data cable/port.
- Old code: transfer did not finish or reached the wrong kit.
- Flashing: wait; never unplug.
- Unexpected R initials/firmware: quarantine; do not reassign casually.



W4 - Prepared NeoPixel Preview

Run and explain first. Code editing is optional after the core pass.

Finish line

Run and explain DIN, data direction, two power sources, shared ground, and A/B/A+B behavior.

Say

P0 sends data into DIN. Positive power stays separate. Grounds join. Students never rebuild the harness.

Before students

- ☐ Reliability-test every 10-12-pixel mentor-built system, harness, resistor, capacitor, and holder.
- ☐ Load A effect, B effect, A+B off, brightness 30-60; match every label.
- ☐ Inspect arrows, connector, heat-shrink, wire, coating, pixels, and batteries.
- ☐ Stage enlarged diagram, whole-system swap, and backup video. Students do not wear the system.

Exact 25-minute flow

Min.	Student action	Mentor gate
0-3	Both holders off; match labels; inspect	Reject damage/mismatch
3-7	Trace P0 -> resistor -> DIN + power	Check arrows/separate positives/shared ground
7-11	Pixels on, then micro:bit; test A/B/A+B	Observe every pixel and off state
11-15	Observe pixel 0, direction, color, brightness	Switch controller/tracer roles
15-19	Optional one color/pause change	Core understanding stays first
19-22	Save photo + explanation	Use whole-system backup, not rewire
22-25	A+B; micro:bit off; pixels off	Verify both supplies off + inventory

Show me

Point to DIN, pixel 0, both power sources, shared ground, and separate positive supplies.

Stop

Warmth, odor, bare wire, torn coating, damaged pixel, unstable flicker, or any request to alter the harness.

Quick fixes

- Dark: check pixel power, connector seating, and DIN direction; do not rewire.
- micro:bit works/pixels dark: check separate pixel supply.
- Flicker/random color: both off; technical mentor checks prepared harness.
- Partial strip: confirm pixel count; mark failure and swap whole system.



R1 + R2 - Identity, Reset, and Input

Teams stay with one numbered kit; partners switch roles each transition.

Roles

Driver operates. Navigator reads the card, checks ports and safety, predicts, and records. Both partners explain. Switch roles at each transition.

Challenge 1 - 9:51-10:16

Min.	Team action	Mentor checkpoint
0-4	Name Hummingbird, R micro:bit, BirdBlox, and power-off rule	Correct system language
4-8	Match kit, iPad, three-word name/initials, and ports	All labels agree
8-14	Power on; assigned device only; green dot; run R1	Wrong-device command stopped
14-20	SAFE RESET; switch roles; repeat start/reset	Both partners complete sequence
20-25	Save label photo + reset explanation	Evidence box cleared

SAFE RESET

1. LED to 0%. 2. Servo to marked ready angle. 3. Stop program. 4. Disconnect BirdBlox. 5. Switch kit power off.

Challenge 2 - 10:19-10:44

Min.	Team action	Mentor checkpoint
0-3	Input reports information; code compares and decides	Sensing is not acting
3-8	Reconnect; identify distance sensor + Sensor 1	Correct device, type, port
8-15	Record far -> near -> far; change only distance	Target centered; three real values
15-20	Choose threshold between observed ranges; predict trigger	No copied/universal threshold
20-25	Explain room/placement effect; save evidence; reset	Value changes for a reason



R3 + R4 - Outputs and Interaction

Use only identical prepared boards and mentor-marked servo angles.

Challenge 3 - 10:47-11:12

Min.	Team action	Mentor checkpoint
0-4	Identify LED 1, Servo 1, ports, and motion zone	Rig secured; roles switched
4-10	Run LED at 0%, 50%, 100%, then 0%	Correct port and safe-off
10-16	Run servo only between two marked safe angles	Hands clear; stop strain/buzzing
16-21	Predict; run LED + servo together; switch explainer	Two outputs observed
21-25	Save screenshot/note; SAFE RESET	LED 0%; servo ready

Say before Challenge 4

IF the object is closer than our measured threshold, THEN the light turns on AND the pointer moves; OTHERWISE the light turns off AND the pointer returns to ready.

Challenge 4 - 11:15-11:40

Min.	Team action	Mentor checkpoint
0-4	Say full IF / THEN / AND / OTHERWISE rule	Input, decision, two outputs, reset named
4-10	Load R4; confirm threshold, ports, safe angles	No rewiring/unapproved angles
10-17	Test far -> near -> far; predict + record; switch role	Three controlled trials
17-21	Change one approved variable; retest same condition	One-variable discipline
21-25	Save screenshot/media; explain; SAFE RESET	Evidence proves full interaction

11:40-11:45

Verify four evidence boxes, both names/contributions, SAFE RESET, disconnect/off, and complete numbered kit inventory.



Safe Rescue and Final Check

Change one thing. Protect the timed learning. Swap a whole prepared set when needed.

Observed	Quick safe check	Do not
Wrong robot moves	Stop both; disconnect/off; verify kit/name/initials; reconnect one at a time	Do not keep testing
Name missing / red dot	Correct R board, power, Bluetooth, no other iPad connected	Do not choose random device or reload firmware
Sensor frozen	Reset/off; Sensor 1 + sensor type; clear view; known-good swap	Do not change several settings
LED dark	Reset/off; LED 1 + brightness above 0%; known-good swap	Do not move prepared wiring
Servo silent	Reset/off; Servo 1 + prepared connection; known-good swap	Do not force shaft
Servo buzzes/stalls	Power off immediately; clear obstruction; technical lead	Do not continue powered
Always true/false	View real values; threshold between near/far; test parts separately	Do not guess threshold

Universal stop conditions

Power off before touching hardware. Stop for heat, odor, loose battery, damaged insulation, pinched wire, unstable rig, wrong device, buzzing/stall, forced motion, or an output that will not stop.

Final kit count

- ☐ Controller
- ☐ R micro:bit
- ☐ Assigned iPad
- ☐ Power pack
- ☐ Distance sensor
- ☐ LED
- ☐ Position servo + pointer
- ☐ Labeled tray parts
- ☐ Saved BirdBlox file + evidence



Needle, Tool, and Kit Count

Print page 11 twice for needle stations; use the lower table for numbered kit issues.

Station: _____ Lead: _____ Date: _____

Checkpoint	Needles out	Needles in	Snips/tools	Power off?	Initials / note
Opening				Y / N	
10:16				Y / N	
10:44				Y / N	
11:12				Y / N	
11:40				Y / N	
Final 11:45				Y / N	

Inspect / missing record

Time	Station / kit	Item or symptom	Safe state	Owner / next action

Release rule

Every needle is returned separately; every powered system is off; every numbered kit is complete or explicitly labeled INSPECT / MISSING with an owner.